Republic of the Philippines
BICOL UNIVERSITY POLANGUI
COMPUTER STUDIES DEPARTMENT
Polangui, Albay

WEB SYSTEMS AND TECHNOLOGIES

PHASE 1: FOUNDATION OF FINAL PROJECT

BU STUDENT ACADEMIC TASK AND DEADLINE MANAGEMENT SYSTEM

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Group 5

BSIS-2B

Submitted to:
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Instructor

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Project Proposal

Project Title

BU Student Academic Task and Deadline Management System

Introduction

Managing school activities can sometimes be difficult, especially when several assignments, projects, and deadlines need to be completed at the same time. Students may forget important tasks or have difficulty keeping track of their progress when they rely only on notebooks, messages, or memory. Because of this, our group came up with the idea of developing the BU Student Academic Task and Deadline Management System, a web application that helps Bicol University students organize their academic responsibilities in one place. The system will give students a simple way to record their school tasks, keep an eye on deadlines, and see which requirements are already done and which ones still need attention.

System Purpose

The purpose of the BU Student Academic Task and Deadline Management System is to help students manage their coursework and meet deadlines more easily. It will allow users to add academic tasks, write descriptions, set due dates and priorities, group tasks by subject or course, and update their progress. Through its dashboard, users can review their pending, ongoing, and completed tasks, along with near or overdue deadlines. Our group also aims to apply what we have learned in Web Systems and Technologies by developing a functional web application using HTML, CSS, Bootstrap, JavaScript, Node.js, Express, MongoDB, and Mongoose.

Target Users

- BU students who need to track assignments, projects, quizzes, and other academic deadlines.
- Students who want to organize their subjects and manage their overall school workload.
- Administrators who need to manage accounts and oversee appropriate system activity.

Objectives

- Provide a simple, understandable interface for managing academic tasks and deadlines.

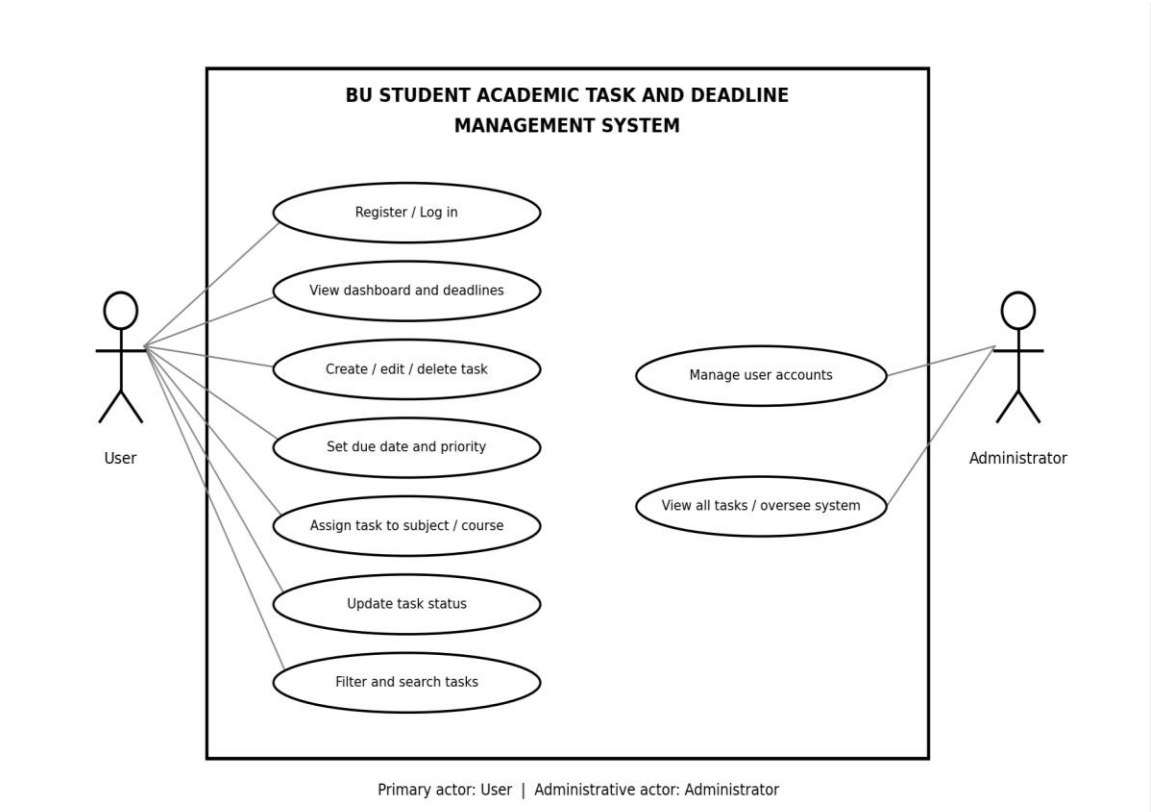
- Help students prioritize schoolwork and monitor task progress and upcoming deadlines.
- Store user and task information in a database so it can be retrieved and updated.
- Practice full-stack development, database modeling, version control, and teamwork.

Key Features

- User registration and login with protected account information.
- Create, view, edit, and delete personal academic tasks.
- Set a task title, description, due date, priority, status, and optional subject or course.
- Filter or search tasks by status, priority, subject or course, or keyword.
- Dashboard showing tasks, their current status, and upcoming or overdue deadlines.
- Administrator account-management and basic oversight features.

Use Case Diagram (UML)

The diagram shows the main actions available to a regular user and an administrator. A regular user manages their own academic tasks and deadlines, while an administrator has additional account-management and oversight responsibilities.



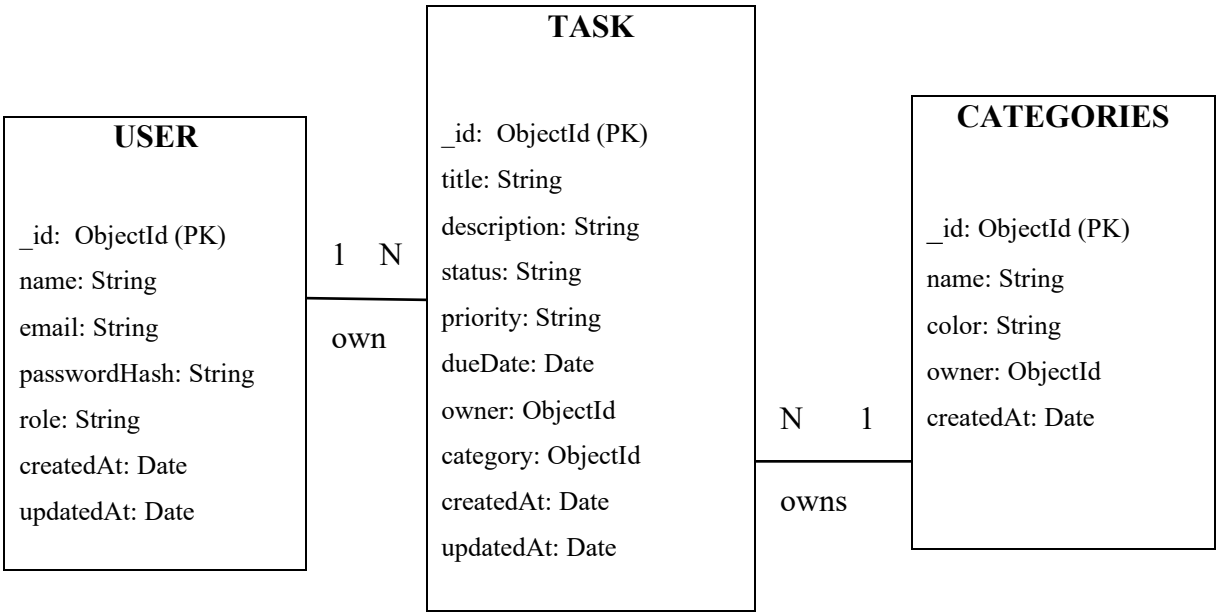
Use Case Notes

- **User:** registers or logs in, views the dashboard and upcoming deadlines, creates/edits/deletes tasks, sets due dates and priorities, assigns tasks to a subject or course, updates status, and searches or filters tasks.
- **Administrator**: manages user accounts and oversees system activity. Administrative actions should be restricted to authorized accounts.
- **Security rule**: a regular user must only access or modify their own tasks; the backend must verify ownership on every task operation.

The Use Case Diagram (UML) was created using draw.io (diagrams.net) to illustrate the interactions between users and the BU Student Academic Task and Deadline Management System, including the main functions that users can perform within the application.

Entity-Relationship Diagram (MongoDB Style)

The proposed design uses three collections: Users, Tasks, and Categories. MongoDB stores documents in collections; ObjectId references connect documents where needed. The category relationship is optional so tasks can still exist without a category. In this system, a category usually stands for a subject or course (for example, Web Systems and Technologies), which lets students group their academic tasks.



Conceptual MongoDB model: each box represents a collection; ObjectId references connect related documents.

Collection and Field Definitions

Collection	Field	Data type	Notes
Users	_id	ObjectId	Primary identifier
Users	name	String	User's display name
Users	email	String	Required; unique; used for login
Users	passwordHash	String	Store a hash, never a plain-text password
Users	role	String	Allowed values: user or admin
Users	createdAt / updatedAt	Date	Managed timestamps
Tasks	_id	ObjectId	Primary identifier
Tasks	title	String	Required task title
Tasks	description	String	Optional details
Tasks	status	String	To Do, In Progress, Completed
Tasks	priority	String	Low, Medium, High
Tasks	dueDate	Date	Required deadline
Tasks	owner	ObjectId	Reference to Users._id; required
Tasks	category	ObjectId	Optional reference to Categories._id
Tasks	createdAt / updatedAt	Date	Managed timestamps
Categories	_id	ObjectId	Primary identifier
Categories	name	String	Subject or course name
Categories	color	String	Optional display color
Categories	owner	ObjectId	Reference to the user who owns the category
Categories	createdAt	Date	Creation timestamp

Relationships and Constraints

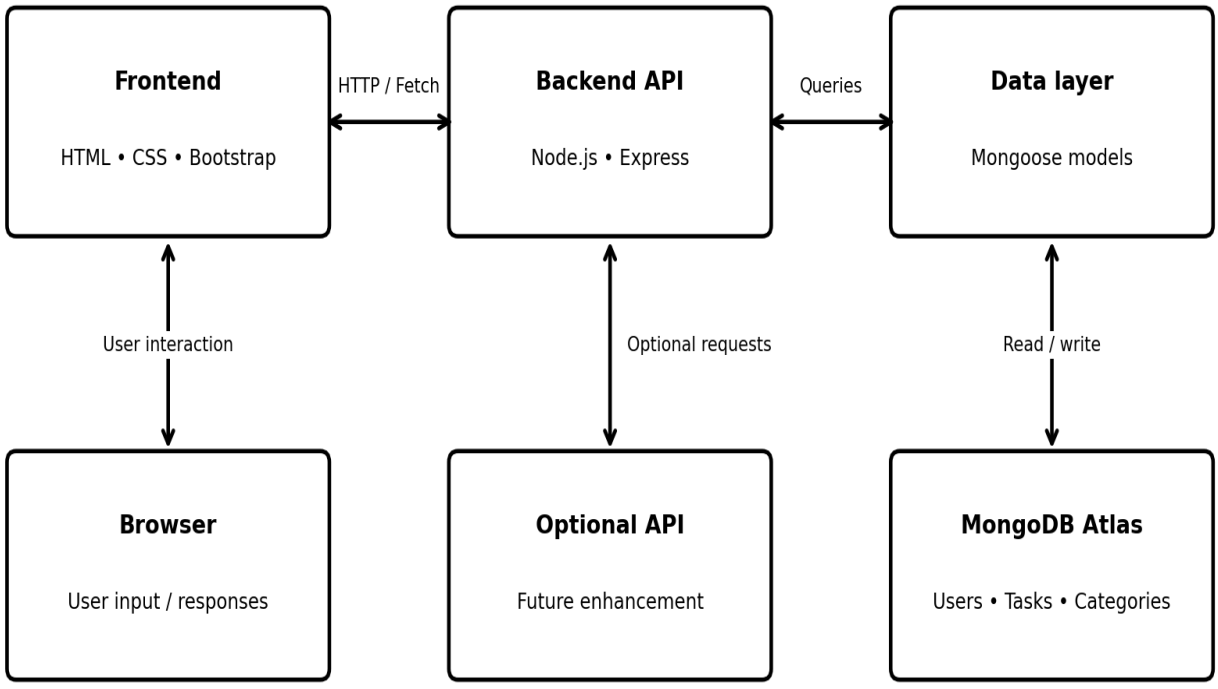
- One User can own many Tasks (one-to-many). Each Task has one owner.
- One User can create many Categories (one-to-many).

- A Task may reference one Category, such as the subject or course it belongs to; category assignment is optional.
- Email addresses must be unique. Required fields (including the task title and due date) and allowed status/priority values must be validated by the backend.
- Use Mongoose schema validation and timestamps. Add indexes where useful, including a unique index on email and indexes on task owner, status, and due date.

System Architecture Diagram

A system architecture diagram is selected instead of a DFD. It shows how the browser-based frontend communicates with the Express backend, which uses Mongoose to access MongoDB Atlas. The optional external API box represents a possible future integration and is not required for the first version.

BU STUDENT ACADEMIC TASK AND DEADLINE MANAGEMENT SYSTEM
HIGH-LEVEL ARCHITECTURE



Authentication and validation are handled by the backend; database credentials stay on the server.

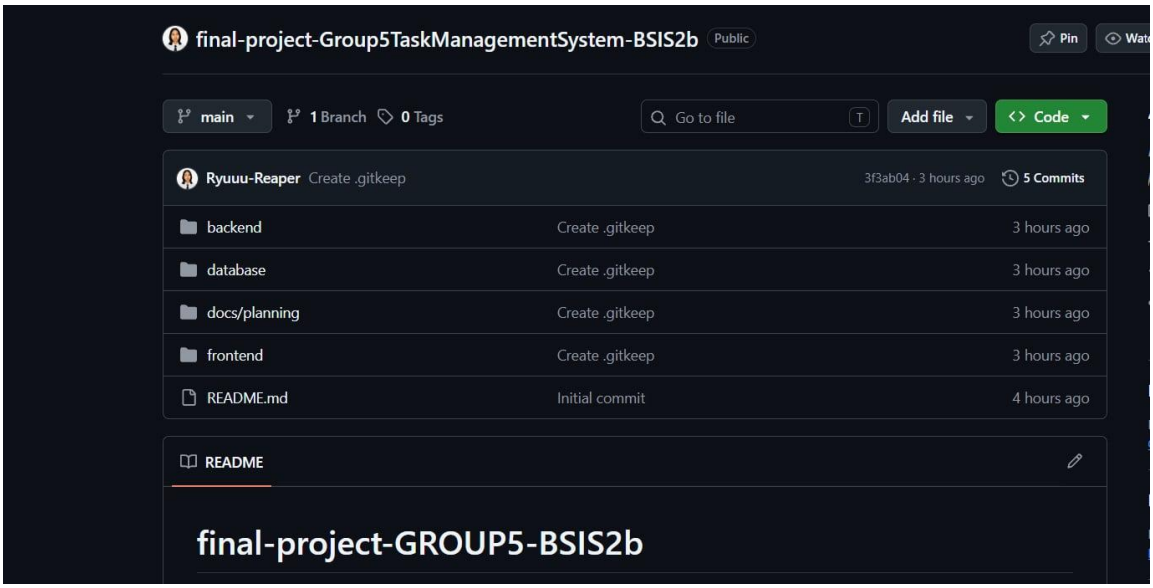
Request Flow

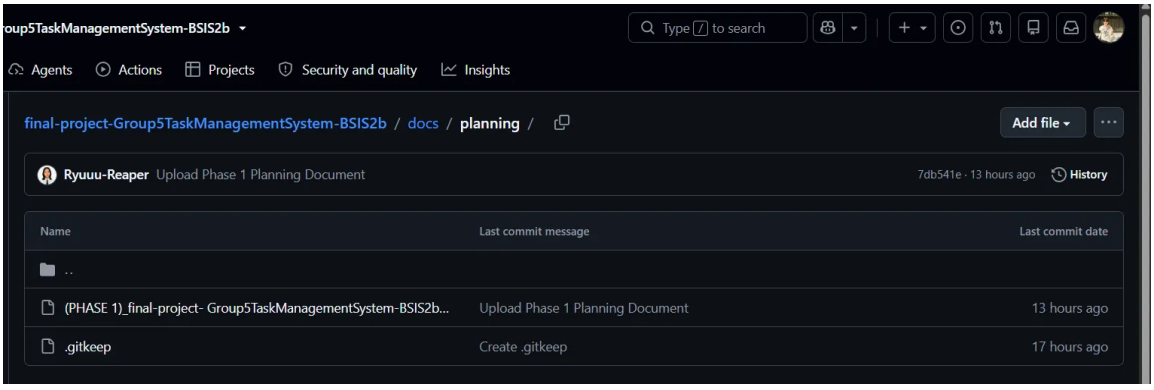
1. The user interacts with a page in the browser, such as the login page or task dashboard.
2. Frontend JavaScript sends an HTTP request to an Express API route using Fetch API.
3. The backend validates the input, checks authentication and task ownership, and calls the appropriate Mongoose model.
4. Mongoose reads or updates the relevant MongoDB Atlas document.
5. The server returns a JSON response. The front end displays the result or an understandable error message.

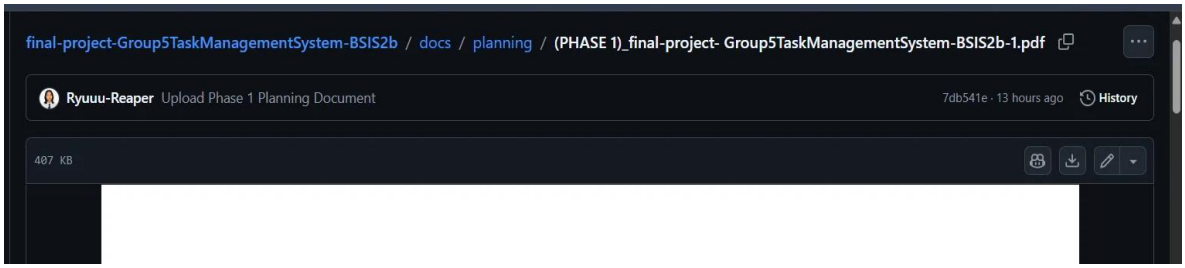
GitHub Repository Setup

Repository name: final-project-Group5TaskManagementSystem-BSIS2b

Repository URL: https://github.com/Ryuuu-Reaper/final-project-Group5TaskManagementSystem-BSIS2b







Personal Role Declaration

Member	Assigned role	Main responsibilities
Bea Roshan O. Pabilona	Project Manager and Documentation Officer	My assigned role is Project Manager, and I will also coordinate the project documentation because our group has five members. I will help the team understand the requirements, divide the work fairly, monitor deadlines, and communicate with our professor when clarification is needed. I will check that the proposal, diagrams, and repository structure are complete and consistent before submission. I also commit to listening to my teammates, helping resolve problems, and making sure everyone contributes to the project.
Eleanor Aimee O. Reantaso	Front-End Developer/ Front-End Manager	As the Front-End Manager, I am responsible for leading the development and design of the user interface for our system. I commit to creating a user-friendly, organized, and responsive interface that allows users to create, manage, and track their tasks easily. I will also coordinate with my groupmates to ensure that the front end works properly with the authentication and data persistence features. I will do my best to contribute my skills, communicate with the team, and complete my assigned tasks on time.

Christian A. Balauro	Backend Developer/ Back End-Manager	As the Back End-Manager my assigned role in this project is the back-end developer. I am responsible for handling the system's logic, processing data, and making sure the functions work properly. I commit to completing my assigned tasks, testing the system, and working with my groupmates to help make our project successful.
Simranjit S. Singh	Database Manager	As the database manager of our project, I commit to organizing and maintaining the database, ensuring that all task and user information is stored accurately and securely. I will work closely with my groupmates, complete my responsibilities on time, and contribute to the success of our project.
Karl James M. Matias	GitHub Manager	As the GitHub Manager, my role is to manage our project repository, organize files, and ensure that all project updates are properly saved and uploaded to GitHub. I commit to keeping our repository organized, tracking changes, and making sure that every member can access the necessary project files. I will also help resolve GitHub-related issues and ensure that our system is properly documented and updated. Through this role, I will contribute to the successful completion of our project by maintaining an organized and reliable workspace for the team.
All members	Testing and debugging	Test the features assigned to each member, report reproducible issues, check fixes, and verify that integrated features work together.

MEMBER SIGNATURES

BEA ROSHAN O. PABILONA
Project Manager and Documentation Officer

ELEANOR AIMEE O. REANTASO
Front-End Developer/ Front-End Manager

CHRISTIAN A. BALAURO
Backend Developer/ Back End-Manager

SIMRANJIT S. SINGH
Database Manager

KARL JAMES M. MATIAS
GitHub Manager